

2082

Bachelor of Science in Computer Science and Information Technology

Course Title: **Computer Networks**

Code No: **CSC263** Semester: **4th Semester**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Section A

1. What are the advantages of layered architecture in network communication? Explain TCP/IP protocol stack in detail.
2. What are the services provided by Transport Layer? Explain working principle of Distance Vector Routing Protocol.
3. Differentiate between IPv4 and IPv6 Protocol in brief. Explain Cyclic Redundancy Check with suitable numerical example.

Section B

Attempt any Eight questions[8×5=40]

4. How connection is established in TCP protocol using 3-way handshaking? Explain.
5. Write subnet ID, broadcast address and address range of each subnet if you divide a class C network 193.168.2.0/24 in 8 different subnets. What is the new subnet mask?
6. What is the major purpose of DNS? Explain how DNS resolve domain name to IP address.
7. Discuss different guided transmission medium in brief.
8. Briefly explain Go-Back-N ARQ and Selective Reject ARQ protocols.
9. What is HTTP protocol? Explain HTTP Request and HTTP Response message in brief.
10. What is virtual circuit network? Explain ATM as a virtual circuit wide area network.
11. What is software defined network? Explain in brief.
12. Write short notes:a. Leaky Bucketb. FTP Protocol